Practical No – 5

Working with Attributes, Terrain Data

A.Working with attributes

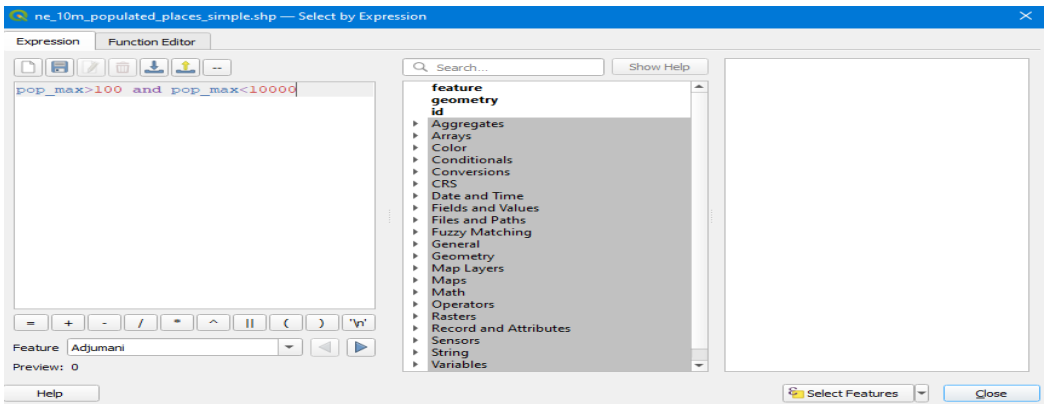
- Start a new project.
- Go to Layer→Add Layer→Add Vector Layer
- Select **"\GIS_Workshop\Practical_04\A\Data\ne_10m_populated_places_simple.zip"**

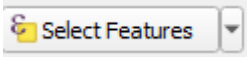
- Right click on Layer Panel→Open Attribute Table.
- Explore various attributes and their values in the attribute table.
- To find the place with maximum population click on **"pop_max"** file.

ne_10m_populated_places_simple.shp — Features Total: 7322, Filtered: 7322, Selected: 0

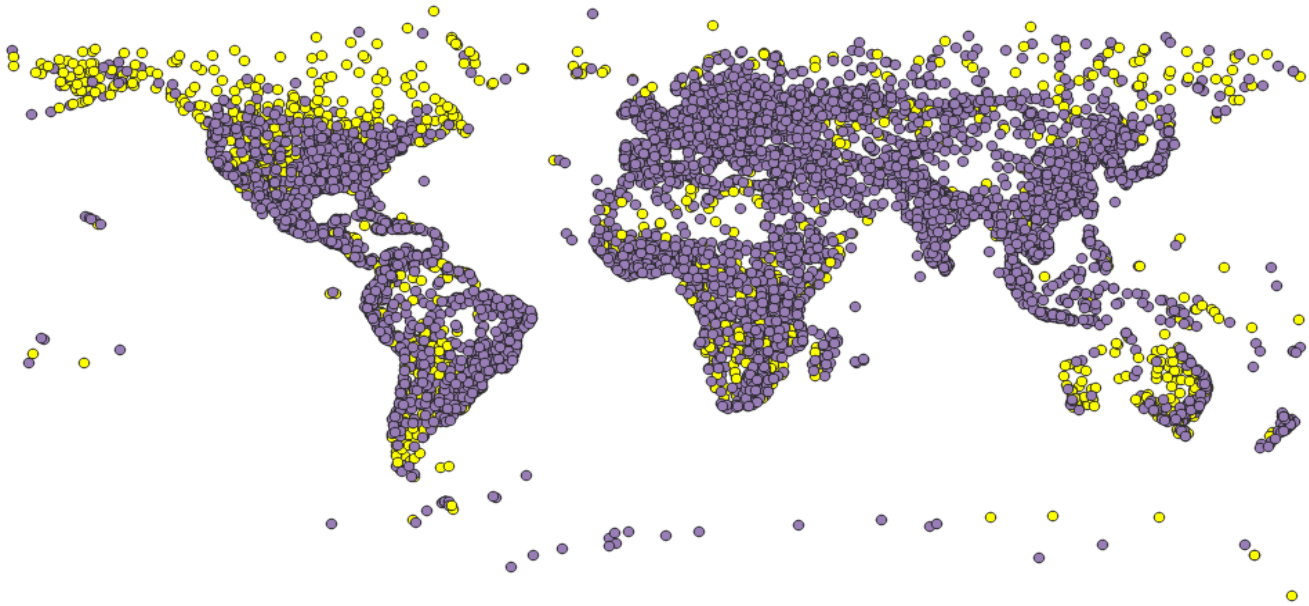
	longitude	changed	namediff	diffnote	pop_max	pop_min	pop_other	rank_max
1	-57.84000247340	4.00000000000	1	Added missing ...	21714	21714	0	7
2	-56.90099656020	4.00000000000	1	Added missing ...	21093	21093	0	7
3	-58.30399747190	4.00000000000	1	Added missing ...	23279	23279	0	7
4	-56.28400149320	4.00000000000	1	Added missing ...	19698	19698	0	6
5	-56.21499844800	4.00000000000	1	Added missing ...	32234	32234	0	7

- On clicking the select feature using expression button the following window will appear.

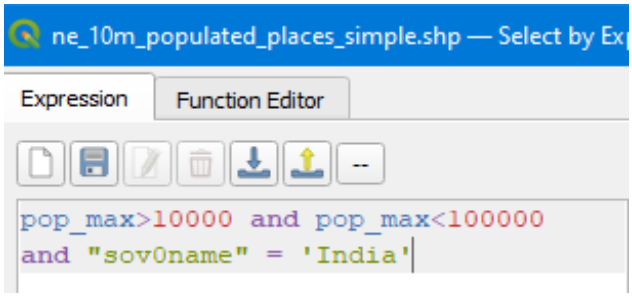


- Enter pop_max>100 and pop_max<10000 and click 

button to get all the places with population between 100 and 10000.
- The places matching the criteria will appear in different colour.



- Different queries can be performed using the dataset.
- Try this

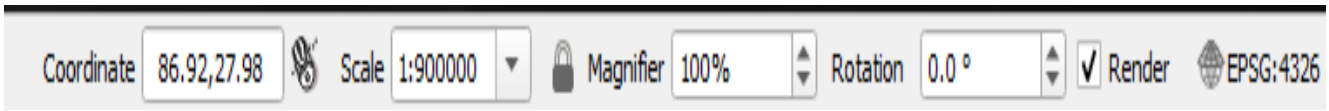


- Use the deselect button to deselect the feature to be rendered in original colour.

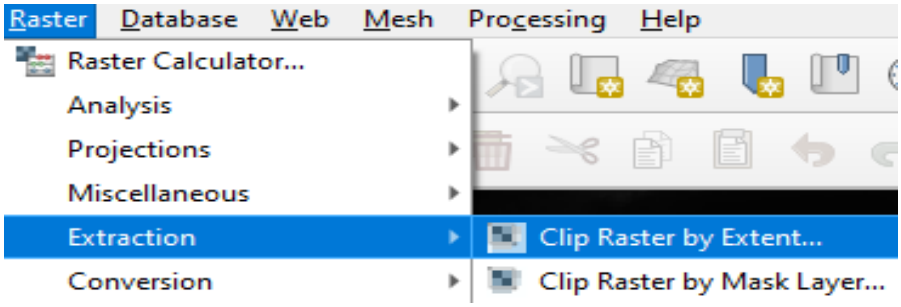
B.Terrain Data and Hill Shade Analysis

- Go to Layer→Add Raster Layer→Select **"10n060e_20101117_gmted_mea300.tif"**, from data folder.

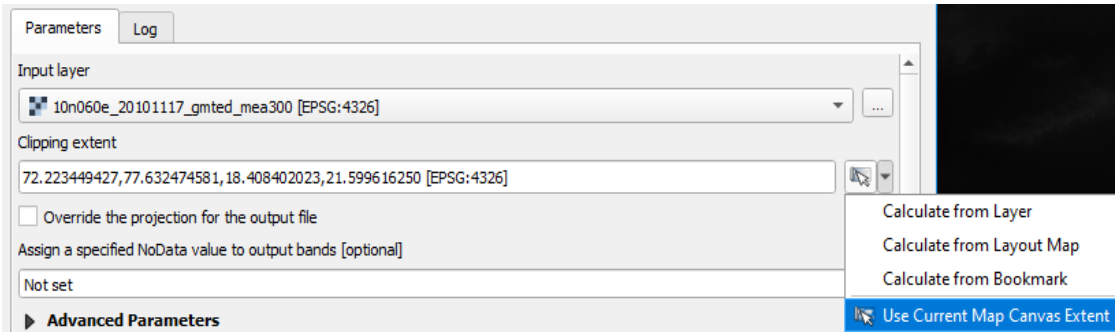
- The lower altitude regions are shown using dark color and higher using light shade as seen on top region containing Himalaya and Mt Everest.
- Enter 86.92, 27.98 in the coordinate field, scale 1:900000 and magnifier 100% at the bottom of QGIS.



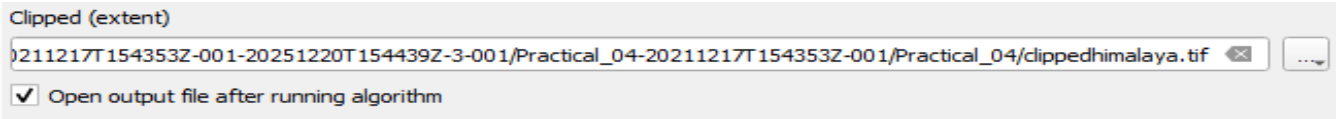
- Press enter the view port will be centred on Himalaya Region.
- Crop the raster layer only for the region under study.
- Go to Raster→Extraction→Clip Raster by Extend



- Select the clipping area by selecting the option **Use Canvas Extends**



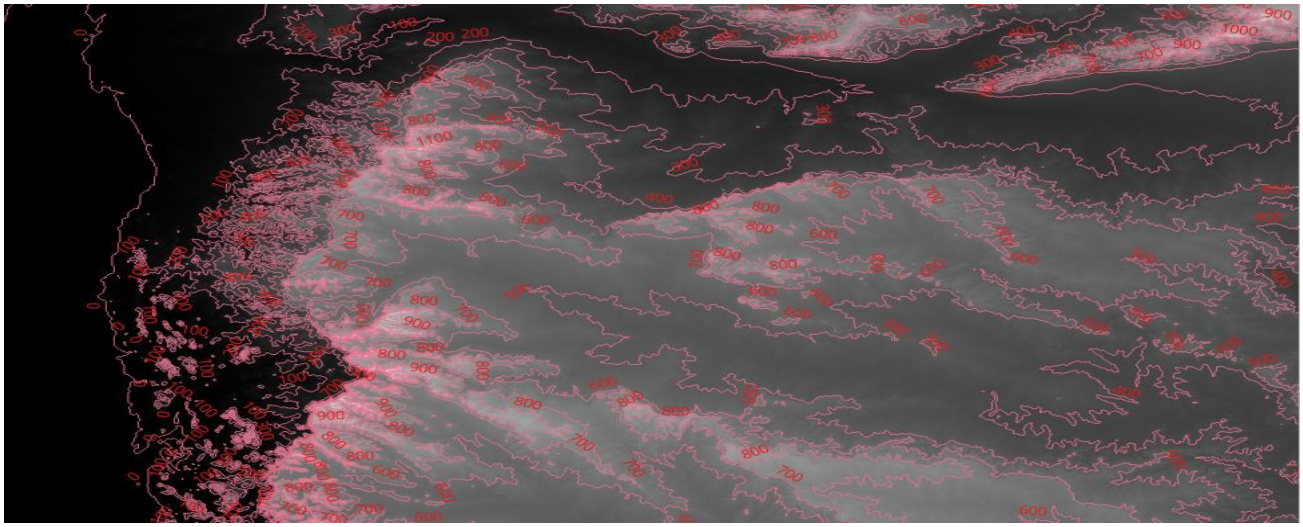
- Select the location and file name for storing clipped raster layer.



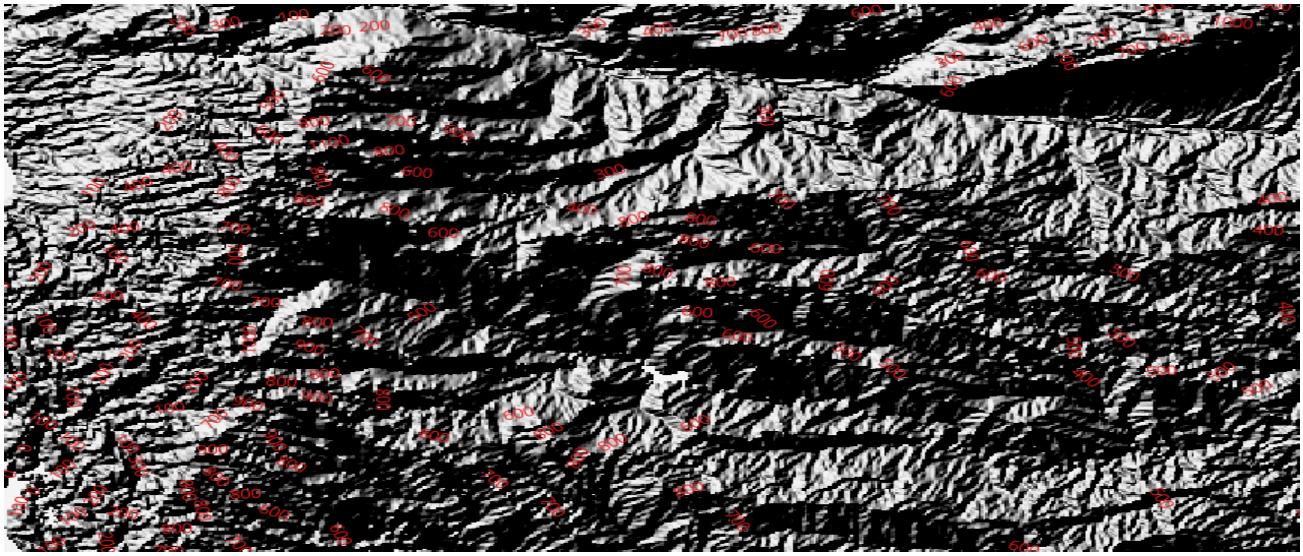
- Press RUN
- Deselect the original layer and keep the clipped one.
- The clipped raster layer is representing altitude are from 103 Meters.
- Counter lines are the lines on a map joining points of equal height above or below sea level.
- To derive counter lines from given raster.
- Go to Raster→Extraction→Contour.

Select the input raster layer name. Set contour interval 100.00 meters

- The contour layer will appear like this



- Label the layer using "ELEV" field and set appropriate symbols for line.

For Hill Shade surface analysis

- Go to Plugin→Install Georeferencer GADL.
- After successful installation of plugin go to Raster→Analysis→Hill Shade